THE CLIMATE OF MADEIRA.

BY M. C. GRABHAM, M.D., F.R.C.P.

ALTHOUGH it must appear to nearly everyone who heard, or has since read an account of, the recent discussion at Cork on the therapeutic value of cold mountain air in phthisis, that the president of the section exercised a wise discretion in withholding the expression of any concurrence of opinion with the views variously stated, I, for one, do not hesitate to state my frank agreement with Dr. Bennet in ascribing, to a great extent, whatever immunity certain districts may enjoy—if there is any exemption—from phthisis, to an open-air mode of life, hygienic precautions, and a moderately equable climate.

For whatever separate share of success diminished pressure, cold, altitude, or antiseptic air may ultimately obtain, no such group of circumstances can yet show any claim to pre-eminence in the treatment or prevention of phthisis; whilst, even in London itself, we often find that the poor man taken suddenly from his home of sickness, filth, and starvation to the comfort and care of hospital treatment shows a measure of benefit quite comparable to that which the rich patient, pent up in cold damp weather—the comforts of an English home notwithstanding—has to seek in the warmth and equability of Mentone or Madeira. The advantages of treatment are relative up to a certain point, and then certain favourable circumstances show a specific value; but beyond this we are all working to find out whether the conditions of absolute security exist. At present we have not got very far.

The purpose of the present communication, however, is more immediately practical, for I gather that there is an unusually widespread disposition on the part of those who spend their winters abroad to go *presently* to Madeira, presumably because Madeira was remarkably and singularly exempt from severe weather last winter, and also because the present English summer proves to be a season of protracted rain and uncertainty. Some disappointment will be saved, I therefore think, if I may state that, although the Quintas (villas) on the mountain slopes of Funchal offer a charming perennial climate of moderate uniformity, it is unwise for invalids, as a general rule, to establish themselves in the town of Funchal until the long period of summer weather has been broken by autumnal rain.

Moreover, notwithstanding the excellence of the Funchal hotels, I am bound to say that the Quintas offer at all times certain advantages of greater value—viz., their situation amidst beautiful scenery; the calmness and purity of their atmosphere; and their gardens, in which a very delicate person may sit out of doors for many hours almost every day in the year. Invalids, moreover, get on better when they do not congregate together; they breathe a purer air; they suffer less from diarrhœa; and they do not acquire the morbid and depressing habit of comparing symptoms.

From Davos I expected to hear much more than has been said about the stillness of the atmosphere, and also, speaking from Madeira, something about a marked immunity from atonic diarrhœa; and I am not disposed meanwhile to weigh a vague, speculative quality of antiseptic air against the tried advantages of open-air life, mild equability, and favourable hygienic conditions.

It may not be out of place here to mention, in connexion with the lull which frequent changes of air often bring about in chronic ulcerative phthisis, the recent development of steam communication between Lisbon, Madeira, and the Western Islands, the latter as yet little visited by English people, though rich in natural beauty and active volcanic phenomena. The Azores have a somewhat humid climate, but there is much fine weather and sufficient solar heat to ripen many of our finest fruits in midwinter. By and by I will offer a detailed description of the Azores, their sulphur baths, and other important points of interest. At present I only wish to say that the steamer alluded to threads its way through the Western Archipelago, stopping a few hours at each island, and that the tour is becoming a common incident in the winter sojourn at Madeira.

Alfred-place, South Kensington.

THE DIAGNOSIS OF VARIOLA FROM VARICELLA VERA.

BY M. D. MAKUNA, L.R.C.P. LOND., &c.,
MEDICAL SUPERINTENDENT, FULHAM HOSPITAL.

WITHIN the last seven months thirty-six cases of varicella have been brought to my knowledge. Thirty-three of these were mistaken for those of variola. Sixteen were admitted to the hospital; they were either vaccinated or revaccinated, and did not take small-pox. Three cases occurred in the hospital, caused by those admitted through misadventure. Last year I had two cases of varicella. In one the revaccination was unsuccessful, but, as the patient was well vaccinated, he took variola in a mild form; another was an unvaccinated child, who fortunately got over it. Observing that such a mistake is on the increase, as chicken-pox is prevalent in the west of the metropolis, and the long run of the epidemic of small-pox has not yet ceased, I take this opportunity of indicating the way in which a diagnosis may be arrived at, and thus prevent recurrence of such mistakes as far as it is possible. I need hardly add that it is of paramount importance to do so before advising the removal of patients to small-pox hospitals.

Varicella is not known to be fatal, and is a very mild disorder; while variola, apart from its high fatality, deprives many a one of comely complexion, and mars the beauty—a sad loss most keenly resented by the possessor in the present age of refinement. We all know that the period of incubation, or the pre-eruptive stage, is nearly the same in both complaints. It does not vary so much in variola as it does in varicella. In my four cases of varicella the period of incubation was respectively seventeen, thirteen, twelve, and nine days. Dr. Gee gives it as about a fortnight. In Dr. Murchison's case it was eleven days. Dr. W. Squire gives it as between ten and twelve days. M. Trousseau and Mr. Hutchinson each state it to be from four to seventeen days. Dr. Thomas, of Leipsic, gives it as from thirteen to seventeen days; and Mr. Erasmus Wilson, from four and seven to fourteen days. In Dr. Steiner's eight successful cases of inoculation of varicella the period was eight days. The pre-eruptive stage in variola is stated generally to be fourteen days. From my observations I find it to be between twelve and fourteen days; but it varies from nine to twenty-one days. Again, the primary fever in both does not give any clue to distinguish them, especially when the prodromata are not marked in variola, as is usual in vaccinated children, in whom the disease is modified and most likely to be mistaken. In variola, it is said that this initiatory fever is characterised by backache, pain in the epigastrium, sickness, and headache. But in many cases the first two symptoms are absent. In many standard works on medicine the invasion stage is said to last forty-eight hours in variola, and twenty-four in varicella. No doubt we do get such a history in many cases; but this statement, I must say, is arbitrary and not precise, so that there is no feature to be relied upon during the course of the pre-eruptive stage to make a certain diagnosis. Curschmann observes, "if we now pass to the consideration of the first stage of variola, we shall find that in many cases, especially of varioloid, the early symptoms are so vague, undetermined, and various, that no diagnosis is possible until the characteristic eruption appears." It would be out of my province to enter into a discussion on the point in this paper, and I forbear making further remarks.

The distinguishing features are made evident soon after the manifestations of the diseases. In variola the eruption is first papular, then vesicular, and finally pustular. In variola varicelloides the pustular stage is abortive. The papules are always felt hard, of nearly the same size whereever they are distinct, and may or may not be felt shotty within the skin at the commencement of their appearance. It takes about three days for a papule to become a vesicle, and between two and three days for a vesicle to turn into a pustule. In the majority of cases the vesicles are umbilicated, especially among the unvaccinated class, but not universally so among the vaccinated. The pathognomonic feature of the vesicles in variola is that they are *always multilocular*, and cannot be emptied of their contents by a single puncture with a pin or a needle. The contents of the vesicles are "plastic and viscous." In varicella the papular stage does not last for many hours. The spots are unequal in size, and on the second and third day they are seen in all stages, as fresh crops come out every morning. The vesicles are superficial, more or less globular, with an undefined areola around them. The fully formed vesicles are *unilocular*, and can be emptied of their contents by single punctures. The lymph is "serous and watery." I have found this a most trustworthy diagnostic feature between the two diseases. Mr. Hutchinson, in his recently published "Lectures in Clinical Surgery," states: "Again I have to express dissent from the popular opinion that the chicken-pox vesicles are always globular. On the contrary, they often, if looked at minutely, show an uneven surface, as if formed by successive effusions. In such cases a single prick does not suffice to remove the whole of the fluid. The assertion that the vesicle is globular, a mere blister with a single cavity, must be taken with limitations." Dr. Tilbury Fox and Mr. Erasmus Wilson, the two eminent dermatologists, state that the vesicles in varicella are unilocular. Dr. Thomas, in his article on Varicella, in Ziemssen's Cyclopædia, states, "When a fresh vesicle is punctured it gives out a clear, or at most very slightly turbid fluid, at first small in amount, but later comparatively considerable, but never sufficient to cause complete collapse of the vesicle; it is evidently not a single vesicular formation, but possesses originally delicate septa, subdividing into single fans, which, nevertheless, can finally run together in whole or part from tearing open their partition walls." So, practically speaking, as he means to say, the fully-formed vesicles are unilocular. Were the vesicles to be multilocular they would be much more irregular in shape than they are observed from the rapid and profuse exudation. They have been described by some as umbilicated, as they are in variola, but the phenomenon is caused by the desiccation of the lymph at the dome of the vesicle sooner than in other parts. The difficulty of evacuating all the spots arises from the fact that all of them are never in one stage. Some are papular or just commencing to vesiculate, while others are vesicles either fully formed and globular, or undergoing desiccation. In this last stage, while tearing through the walls of the vesicles, I have removed clots of dried-up lymph.

The object of this paper has been to draw the attention of the profession to this point, and to the means of arriving at a correct diagnosis between these two disorders, one of which is never fatal, and the other one of the direst and most loathsome legacy of humanity. When there is reliable history of exposure to the infection, the nature of the disease is evident, but unfortunately this is not always to be had.

Fulham.

RADICAL CURE OF HYDROCELE.

BY JOHN D. HILLIS, F.R.C.S.

HYDROCELE of the tunica vaginalis being very commonly met with in tropical climates, it becomes a matter of some practical importance to those practising there to understand what treatment is most likely to effect a radical cure. The injection generally recommended in text-books (composed of one part of tincture of iodine and two or three parts of water) will do this in but a small proportion of the cases operated on, occasioning, in many instances, disappointment and loss of time. Mr. Osborn, writing on the subject, gives the results in his hands of the diluted iodine injection operation as follows:—"Out of 54 cases operated on 19 had been previously injected unsuccessfully. Of these 54 he lost sight of 29, and in 18 of the remaining 25 the hydrocele recurred"—not a very satisfactory record of the operation, even if the 29 lost sight of had been all cured by it.

I have for some time past used for the purpose the *un*diluted tincture of iodine of the British Pharmacopœia, and abandoned, after a fair trial, nearly all the others. I inject a small quantity, from two to four drachms, according to circumstances, and, if possible, at the first tapping; the smaller the quantity injected the safer the operation, and less risk of infiltration. After injection the sac should be well but gently manipulated for some time. The results have been so satisfactory that, simple as the matter may be, and old as the suggestion perhaps is, I do not think it has been sufficiently recognised by teachers as a procedure recommended to be adopted; I therefore bring it forward for the benefit of my junior brethren practising abroad. I have selected the following cases from my note book as illustrating this mode of treatment in persons of different nationalities.

1. E. H——, European, aged twenty-six, a planter; double hydrocele three years. A surgeon injected the right side with the ordinary iodine injection in 1877, but the hydrocele recurred. Tapped the left side in May, 1878, and injected equal parts of tincture of iodine and water, which also failed to effect a cure. On the 17th June, 1878, the right side was again tapped, and this time I injected three drachms of pure tincture of iodine, and manipulated the parts well for some time. Very little pain was experienced. In two days the parts were well swollen, and in ten days that side was cured. Subsequently I performed a similar operation on the other side with an equally satisfactory result. There has been no recurrence of the disease.

2. T. K——, a negro, aged thirty-seven, blacksmith; large hydrocele of right sac, from which over four pints of fluid were evacuated; had been previously unsuccessfully injected with diluted iodine. On the 5th June injected six drachms of pure tincture of iodine. The patient experienced very little inconvenience (a gland slightly enlarged in the neighbourhood), and he was discharged cured on June 16th.

3. Enyek, a Hindoo, aged twenty-five; double hydrocele. Left side previously injected unsuccessfully with the diluted tincture of iodine. On the 5th December, at the Helena Hospital, injected right side with half an ounce of pure tincture of iodine. Discharged cured on the 16th December. Readmitted on the 21st December, and the left side, where there is some chronic enlargement of the testis, injected in the same way, the patient being discharged cured on the 31st December, 1878.

British Guiana.

CASE OF TETANUS FOLLOWING BURNS; RECOVERY.

BY JAMES COULDREY, M.R.C.S.E., &C.

THE following case may be deemed of sufficient interest to be placed on record in THE LANCET:—

George S——, aged thirty-six, a workman at the Trent Iron Company's blast furnaces, had both his feet badly burned by stepping on and sinking into a hot slag-ball on June 22nd, 1878. The skin and subcutaneous tissues were destroyed around the ankles and over the dorsal aspect of both feet, including the toes; the great toe of the left foot was more deeply burned than the rest. The burns were dressed with carron oil, and subsequently poultices were applied to promote separation of the sloughs.

On July 9th he complained of his jaws being stiff, also pain in the small of the back; left foot very painful; wounds healthy and granulating. Ordered a pill containing one grain of opium powder and four grains of extract of hyoscyamus every four hours. Wounds dressed with carbolic acid and zinc ointment.

10th.—Jaws stiff, can just get point of finger between them; bowels open; pulse 70. Wounds dressed as before; to continue pills, and take plenty of fluid nourishment.

13th.—Jaws and back about the same; manages to get pills down; has taken fluid nourishment; twitchings of muscles of left leg noticed; wounds look healthy. Ordered a mixture of bromide of potassium, twenty grains for a dose, every two hours; the pills as before; to have an aperient pill.

14th.—Stiffness of the jaws somewhat better; muscles of the back very hard and tense; notwithstanding aperient medicines the bowels are not moved. To take another aperient in the shape of a draught. In the evening jaws about the same; tongue and skin moist; temperature 99·2°; pulse 80; bowels not moved; small of the back arched and raised from the bed; abdominal and dorsal muscles tense; says he is in great agony. Whilst trying to give him some nourishment he uttered a cry, and was turned to his left side, the body being arched (pleurothotonos); on getting on his back again he had another attack, this time arching his back (opisthotonos). These spasms